

Project Proposal: US Traffic Accident Severity & Risk Modeling

Course: CSGY-6513 Big Data

Section: C

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Project Abstract

Road traffic accidents are a major public safety issue in the United States, resulting in thousands of injuries and fatalities each year. Understanding the factors that contribute to accident severity is critical for improving road safety and implementing preventive strategies. This project proposes a big-data-driven analysis of U.S. traffic accident records from 2016–2023 using distributed computing frameworks.

The dataset contains millions of accident records including geographic location, weather conditions, time of occurrence, and road characteristics. By leveraging Apache Spark and PySpark for large-scale data processing, the project aims to identify patterns associated with high-severity accidents. Machine learning models such as Logistic Regression and Random Forest will be applied to predict accident severity levels and analyze key contributing factors.

The results of this project will provide insights into environmental and temporal conditions that increase accident severity, enabling better risk assessment and supporting data-driven road safety initiatives.

Problem Statement

Traffic accidents occur frequently across the United States and often result in severe injuries, fatalities, and significant economic losses. Accident severity is influenced by multiple factors including weather conditions, road environment, visibility, traffic density, and time of day. Analyzing these factors requires processing large volumes of accident data collected across multiple states and years.

Traditional data processing tools are not efficient when dealing with millions of records. The dataset contains over 7 million records (2–3 GB), requiring distributed computing for scalable and efficient analysis. Big data technologies such as Apache Spark enable scalable processing and analysis of large datasets. By applying distributed data processing and machine learning techniques, it becomes possible to identify patterns that contribute to severe accidents and develop predictive models that can estimate accident severity under different conditions.

This project aims to build a scalable big-data analytics pipeline to analyze traffic accident data and identify key risk factors associated with high-severity accidents.

Objectives

1. Design Scalable Distributed Data Pipeline

Process 3–5 million accident records efficiently using distributed computing frameworks to perform large-scale cleaning, transformation, and analysis.

2. Analyze Severity Trends (2016-2023)

Examine longitudinal trends to identify:

- Year-over-year changes in accident severity rates
- Seasonal and monthly patterns
- Time-of-day risk variations
- Emerging trends in post-pandemic traffic conditions

3. Examine Environmental and Temporal Risk Factors

Quantify the impact of:

- Weather conditions
- Visibility levels
- Temperature and wind speed
- Sunrise/Sunset conditions and nighttime driving patterns

4. Identify High-Risk Geographic Regions

Perform state- and city-level comparisons to detect severity hotspots and regional disparities in accident outcomes.

5. Construct a State Severity Risk Index (SSRI)

Develop a normalized risk index that ranks states based on the proportion of high-severity crashes, enabling meaningful cross-state comparison.

6. Build a Moderate Machine Learning Model

Implement and evaluate classification models (Logistic Regression and Random Forest) to:

- Predict high-severity accidents

- Compare model performance
- Ensure scalability within distributed ML pipelines

7. Interpret Key Risk Factors for Actionable Insights

Analyze feature importance and model coefficients to:

- Identify dominant contributors to high severity
- Translate findings into practical road safety recommendations
- Support data-driven policy and infrastructure improvements

Data Source

Dataset Name: US Accidents Dataset (2016–2023)

Source: Kaggle

Link: <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>

Dataset File Size: 2-3 GB

Approximate Number of Records : 7+ million records

Project Subset: 3- 5 million records

Geographic Coverage: 49 US states

Key features:

Core Severity & Time Attributes

- Start_Time
- Severity Scale(1-4)
- Sunrise_Sunset

Geographic Features

- State
- City
- Latitude
- Longitude

Environmental Conditions

- Weather_Condition
- Temperature
- Visibility
- Wind_Speed

Road & Infrastructure Features

- Junction
- Traffic_Signal

- Crossing
- Stop
- Traffic_Calming

Proposed Technologies

1. **Google Cloud Dataproc:** We will use Dataproc to create and manage an Apache Spark cluster for processing the large accident dataset efficiently.
2. **Apache Spark:** Spark will be used to handle distributed data processing since the dataset contains millions of records.
3. **PySpark & Spark SQL:** These will help us clean the data, engineer new features, and perform large-scale queries and aggregations.
4. **Spark MLlib:** We will use MLlib to build and evaluate classification models such as Logistic Regression and Random Forest for predicting accident severity.
5. **Google Cloud Storage (GCS):** The dataset will be stored in GCS so it can be accessed by the Spark cluster.
6. **Parquet:** We may convert the dataset into Parquet format to improve processing speed and efficiency.
7. **Matplotlib / Seaborn:** These libraries will be used to create visualizations for severity trends, state risk comparisons, and model results.

Programming Language

1. **Python:** Python will be the primary language used throughout the project for data processing, distributed analytics, and machine learning implementation.